

API TESTING CASE STUDY

Professional QA Documentation & API Validation Report

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This document demonstrates a professional API testing workflow including endpoint validation, response verification, authentication testing, and bug reporting practices.

1. Project Overview

This case study demonstrates the API testing process for a Task Management Web Application. The project focused on validating authentication APIs, user management APIs, and task operations APIs to ensure response accuracy, security validation, and reliable backend functionality.

2. Tools & Technologies Used

- Postman
- REST API
- JSON
- Swagger Documentation
- Chrome DevTools
- Linear / Jira
- VS Code

3. API Endpoints Tested

Method	Endpoint	Purpose
POST	/api/login	User Authentication
GET	/api/users	Fetch User List
POST	/api/tasks	Create Task
PUT	/api/profile	Update User Profile
DELETE	/api/task/12	Delete Task

4. Request & Response Sample

Request:

```
{
  "email": "test@gmail.com",
  "password": "123456"
}
```

Response:

```
{
  "status": 200,
  "message": "Login Successful",
  "token": "JWT_TOKEN"
}
```

5. Validation Points

- Status code validation
- Response body validation
- Authentication token validation
- Response time validation
- Data accuracy validation
- Error handling validation
- Negative testing scenarios

6. Bug Findings

BUG-01: Invalid authentication token still returns status code 200 instead of 401.

BUG-02: Empty request body creates user account successfully without validation.

BUG-03: API response time exceeds acceptable threshold during heavy load testing.

7. Test Scenarios

- Verify valid login functionality
- Verify invalid password behavior
- Verify unauthorized access handling
- Verify response schema structure
- Verify task creation functionality
- Verify API error messages
- Verify profile update functionality

8. Conclusion

The API testing process successfully validated authentication workflows, response structures, and backend functionality while identifying several validation and security-related issues. This project demonstrates practical QA testing knowledge and professional API testing documentation practices.